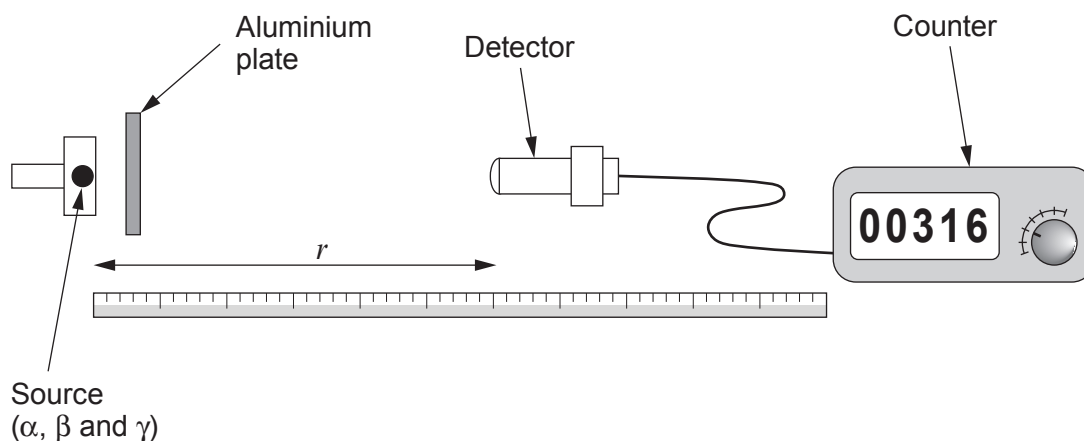


5. A student says that the radiation intensity, I , of gamma rays increases with decreasing distance, r , from a source such that:

$$I \propto \frac{1}{r^n}$$

She wishes to determine the value of n .

The student uses a source of alpha, beta and gamma radiation. She measures the intensity of the rays at set distances from the source by determining the count over intervals of one minute. She also takes measurements without the source at the start and at the end of the experiment.



- (a) (i) Suggest why a thin aluminium plate is placed near the source. [1]

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- (ii) Explain why readings are taken without the source, at the start **and** at the end of the experiment. [2]

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- (b) At each distance, r , the counts were measured over one minute intervals three times. The uncertainty in r is ± 0.005 m. The measurements are recorded in the table.

Complete the table.
Space for calculations.

[2]

Distance, r / m	Count				Uncertainty in mean count
	First reading	Second reading	Third reading	Mean	
without source	19	22	25	22	3
1.000	110	130	136	125	13
0.800	195	165	178		
0.600	270	316	300		
0.400	661	604	651	639	29
without source	20	20	25	22	3

- (c) (i) The student subtracts the mean count without the source from the mean count for each distance to obtain the corrected mean count, N . She also determines the uncertainty in each of these corrected mean counts.

Complete the table.
Space for calculations.

[2]

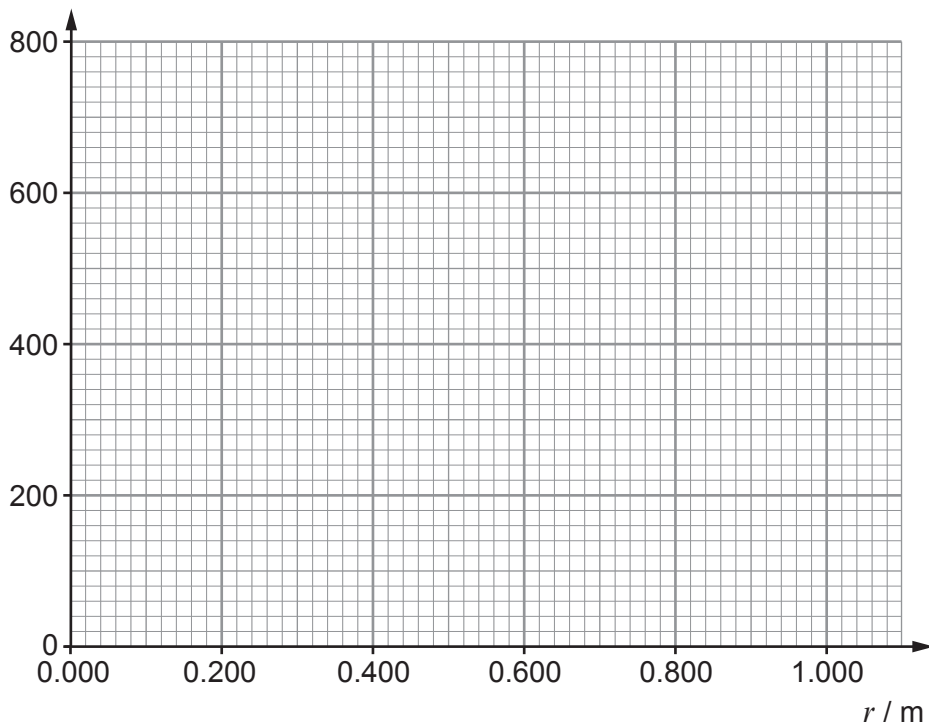
Distance, r / m	Corrected mean count, N	Uncertainty in corrected mean count
1.000	103	16
0.800		
0.600		
0.400		



Examiner
only

- (ii) Plot a graph of the corrected mean count, N , against distance, r . **Error bars are not required.** [3]

Corrected mean count, N



- (iii) The student claims that error bars may be shown on the graph for the corrected mean count, but not the distance. Justify her claim. Calculations are not required. [1]

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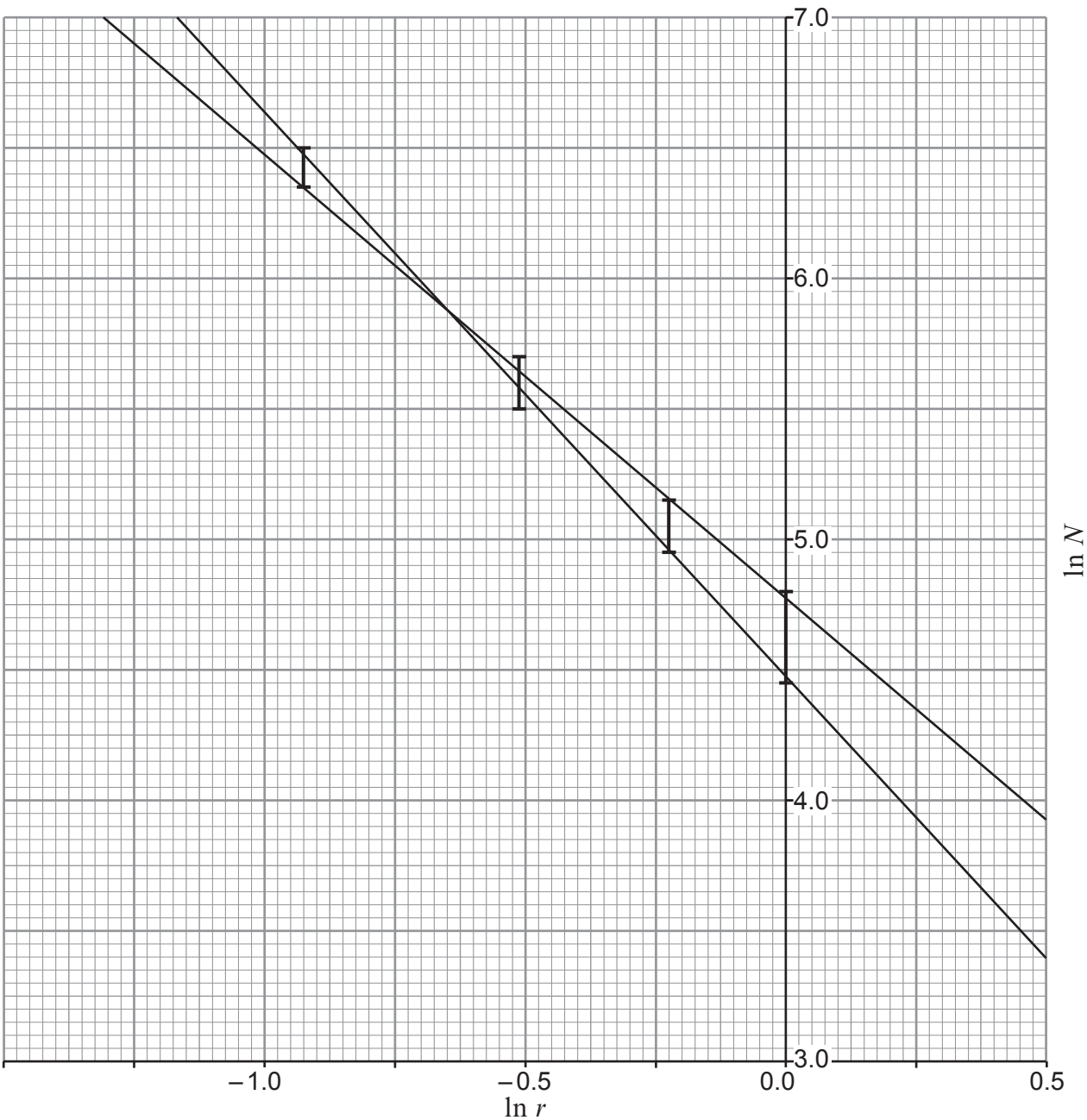
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- (d) Her friend suggests using a logarithmic graph. She calculates the logarithmic values in the table below.

Distance, r / m	$\ln r$	$\ln N$
1.000	0.00	4.63
0.800	-0.22	5.06
0.600	-0.51	5.61
0.400	-0.92	6.42

The data are plotted on the graph.



Examiner
only

For a distance of $r = 1.000\text{ m}$ use the table in part (c) to show that the uncertainty in the logarithmic counts for this distance is approximately 0.15. [3]

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(e) (i) Starting with intensity: $I \propto \frac{1}{r^n}$ justify how the logarithmic graph can be used to determine n . [3]

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(ii) Use the maximum and minimum gradients to determine n and the **absolute** uncertainty in its value. [4]

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